AN2DL - Second Challenge Report
ReLUctant

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1 Introduction

This assignment focuses on the classification of low-magnification Whole Slide Images (WSIs) of human tissue into four distinct molecular subtypes associated with potential diseases. Specifically, the task involves identifying breast cancer subtypes. As better discussed in the experiment section, we followed the pipeline described by Liu et al. [2], adapting their strategies to our dataset.

2 Data analysis and cleaning

Every sample of the given dataset is composed of two distinct components: the image itself and a mask delimiting the border of the cancerous portion of the cell.

From a first analysis of the training set, we noticed that a subset of the images was provided dirty; In particular, they **were compromised** by significant artifacts: these images contained extraneous marker ink or **distinct greenish anomalies** (Shrek's face or green slimes). To avoid noise in the training phase, we implemented a custom filtering algorithm specifically designed to detect and reject contaminated slides automatically.

Another key aspect of the dataset is the **class imbalance**: the distribution of classes (representing the four breast cancer subtypes) were not uniform. Certain subtypes, such as Luminal A and Luminal B, were more prevalent, while Triple Negative samples were relatively underrepresented, as shown in the bar-plot in Figure 1

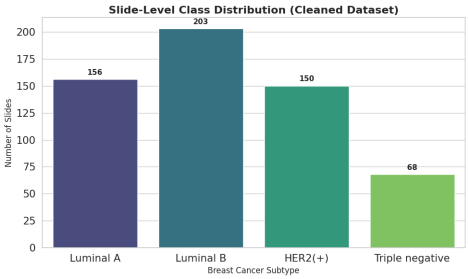

Figure 1: Breast cancer subtypes distribution

3 Method

3.1 Preprocessing Pipeline

3.1.1 Normalization

We initially implemented a standardization pipeline combining

- **Macenko Normalization** [3]
- **Araujo Contrast Stretching** [1]

to mitigate staining variations. However, qualitative analysis suggested that this approach could amplify artifacts in sparse tissue regions, leading to unnatural color saturation. Consequently, we also explored a raw data strategy using standard **Z-score normalization** during the embedding phase. This allowed us to compare the efficacy of the two normalization strategies.

3.1.2 Patches extraction:

We implemented a **mask-guided extraction** pipeline at different resolutions (we tried 224×224 and 256x256), partitioning the dataset at the slide level to strictly **prevent data leakage**. Patches were retained only if the given masks indicated significant tumor content ($\geq 1\%$). To implicitly augment the training data, we utilized a 50% overlap, whereas validation patches were extracted without overlap to ensure unbiased evaluation.

We also experimented with same-size, overlapping training and validation patch extraction to simplify the final *K-Fold Cross Validation* experiments.

3.2 Pipeline architecture

To ensure faster training, we saved to disk the extracted patches and the feature bags generated by the foundation model. This way, we cut training time and optimized training resources.

3.3 Data Augmentation

In the final version, by leveraging the `Phikon` foundation model, we omitted standard geometric augmentations. While earlier experiments with CNN architectures like `ResNet` and `DenseNet` (described in the next sections) necessitated heavy augmentation to achieve generalization, `Phikon` is trained using *Self-Supervised Learning* (**SSL**) objectives. This training explicitly forces the model to learn feature representations that are **invariant** to global view changes, rotations, and color shifts, rendering additional synthetic augmentation redundant.

3.4 Training Strategy

3.4.1 Loss Function

Categorical cross-entropy with class weights computed for minority class loss weighting. We also experimented with different combinations of:

- *Label Smoothing loss:* a custom made categorical cross-entropy extension that also maps class labels to properly-distanced floats rather than equally-distanced integers.
- *Focal loss:* a categorical cross-entropy extension designed to address class imbalance in classification tasks by giving higher weights to hard-to-classify data samples while downweighting samples classified with higher confidence.
- *Contrastive Loss:* a self-supervised auxiliary loss that encourages the model to learn robust, discriminative features by maximizing agreement between differently augmented views of the same slide while pushing apart features from different slides:

3.4.2 Regularization Techniques and optimization

We first used the `AdamW` optimizer during preliminary testing. Then, we evaluated `Lion`. While `Lion` converges faster than the alternative, it is much more unstable.

To adjust to this instability, we found similar performance to `AdamW`, with faster convergence, experimenting with

Hyperparameter	AdamW	Lion
Min LR	10^{-5}	10^{-7}
Dropout	10^{-2}	10^{-2}
L1	10^{-5}	10^{-5}
L2 (WD)	10^{-4}	10^{-4}
LR WarmUp ep.	10	15
Scheduler	ReduceLRoP	CosineAnn

Table 1: Hyperparameter configurations

4 Experiments

4.1 Architecture Evolution

We explored multiple architectures in our progression toward an optimal Multiple Instance Learning (MIL) solution:

1. CNN Baselines: Initially, we experimented with standard CNN architectures (`ResNet18`, `ResNet50`, `DenseNet121`, `EfficientNetB0`) applied directly to the slides. Despite extensive data augmentation, the models lacked generalization capabilities on the *test set* as the performance proved to be slightly better than a four-sided-coin toss.

2. Patch-Based ResNet: We then adopted a patch extraction strategy with `ResNet18` as a feature extractor, followed by aggregation methods. While this improved performance, the model struggled to generalize on the patches.

3. MIL with Attention: We transitioned to a Multiple Instance Learning framework using gated attention mechanisms to aggregate patch-level features. This allowed the model to learn which patches were most relevant for classification.

4. Transformer-Based MIL: Our final architecture combines:

- **Phikon** foundation model (768-dim features) for patch-level feature extraction
- **Extraction of geometric features from masks** using a geometric mask encoder (128-dim features)
- **Transformer encoder** (3 layers, 8 attention heads) for modeling patch relationships
- **Gated attention** mechanism for slide-level aggregation
- **LayerNorm** throughout for training stability with small batch sizes

4.2 Further augmentation

After implementing the MIL pipeline, we experimented with different augmentation and noise injection techniques, both on the extracted patches and on the feature bags extracted from the foundation model.

We experimented[1] with stochastic **CutMix** patch extraction and Feature Bags **MixUp** from same-class and same-slide patches and related features with different degrees of probability; however, the foundation model proved to be robust to these changes, as performance did not improve from these trials.

4.3 K-Fold Cross-Validation

To obtain robust performance estimates, we implemented stratified 5-fold cross-validation at the slide level, ensuring no data leakage between folds. The final submission used an ensemble of all 5 fold models, soft-averaging their predictions to improve robustness and reduce variance.

5 Results

As the challenge progressed, we tested new models and introduced new pipeline, augmentation and training improvements, also following suggestions provided in the *Log Book*.

As Table 2 suggests, further modifications of the MIL Transformer model took us to the final K-fold experiment, which yielded the best performance.

6 Discussion

6.1 Strengths of Our Approach

1. **Foundation Model Leverage:** Using Phikon pre-trained on histopathology provided domain-specific features superior to `ImageNet`-pretrained models.
2. **Advanced/experimental training:** Combination of focal loss, contrastive learning, different slide, patch and feature augmentation techniques addressed both class imbalance and feature quality.
3. **Robust Aggregation:** Transformer + attention mechanism captures global slide context from related patches.
4. **Ensemble Strategy:** K-fold ensemble with uncertainty quantification improved reliability and reduced prediction variance.

6.2 Limitations and Lessons Learned

1. **Augmentation Risk:** Aggressive augmentation can create unrealistic patterns that do not generalize.
2. **Computational Cost:** Full-resolution patch extraction and feature computation required significant GPU resources and time.
3. **Limited Interpretability:** While attention weights provide some insight into important regions, understanding *why* the model makes specific predictions remains challenging.

7 Conclusions

Starting from standard CNN baselines, we moved toward a **MIL framework** to address the specific challenges of high-resolution tissue analysis and class imbalance.

Our final solution effectively integrated the `Phikon` foundation model for **domain-specific** feature extraction with a **Transformer-based** aggregation mechanism, supplemented by geometric features derived from tissue masks. Crucial to our methodology was the implementation of advanced training strategies, including focal loss, to stabilize convergence on underrepresented classes. Experimental results shown in Table 2 demonstrate the potential of this approach.

Future developments should focus on multi-scale patch processing to capture broader contextual information beyond fixed resolutions. Additionally, further work is required to enhance model interpretability, ensuring that the attention mechanisms provide clinically meaningful insights into the morphological features driving the classification.

[1] In a different notebook, called `experiments.ipynb`

References

[1] Teresa Araújo, Guilherme Aresta, Eduardo Castro, José Rouco, Paulo Aguiar, Catarina Eloy, António Polónia, and Aurélio Campilho. Classification of breast cancer histology images using convolutional neural networks. *PLOS ONE*, 12(6):e0177544, 2017.

[2] Hong Liu, Wen-Dong Xu, Zi-Hao Shang, Xiang-Dong Wang, Hai-Yan Zhou, Ke-Wen Ma, Huan Zhou, Jia-Lin Qi, Jia-Rui Jiang, Li-Lan Tan, et al. Breast cancer molecular subtype prediction on pathological images with discriminative patch selection and multi-instance learning. *Frontiers in Oncology*, 12:858453, 2022.

[3] Marc Macenko, Marc Niethammer, James S Marron, David Borland, John T Woosley, Xiaojun Guan, Charles Schmitt, and Nancy E Thomas. A method for normalizing histology slides for quantitative analysis. In *2009 IEEE International Symposium on Biomedical Imaging: From Nano to Macro*, pages 1107–1110. IEEE, 2009.

Appendix

Table 2: Summary of Experimental Results and Model Progression

Experiment Category	**Validation F1** (Mean ± Std)	**Public Test F1** (Best Recorded)
CNN-Based Experiments	0.2315 ± 0.018	0.2343
Pretrained Foundation CNNs	0.2640 ± 0.027	0.2706
Patch-Based Training	0.3310 ± 0.023	0.3038
MIL Transformer Learning	0.3695 ± 0.016	0.3650
K-Fold MIL Transformer	$\mathbf{0.4210 \pm 0.067}$	**0.3963**